higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

AUGUST EXAMINATION

MATHEMATICS N5

2 AUGUST 2016

This marking guideline consists of 10 pages.

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INSTRUCTIONS AND INFORMATION

1. ✓ equals 1 mark.

2. Half marks are not allocated, unless indicated otherwise.

3. Using the wrong formula is a principle error and NO marks will be allocated.

4. Candidates should show ALL formulae and intermediate steps and simplify where possible.

5. ALL final answers must be rounded off to THREE decimals unless indicated otherwise.

6. Questions may be answered in any order, but subsections of questions must be kept together. If subsections are separated, the candidate can be penalised with ONE mark.

7. Where something has been copied incorrectly from the question paper, and the standard of the question is still the same, the candidate will be penalised with ONE mark. If the copying error simplifies the question and makes it easier, the student forfeits the marks.

8. Questions must be answered in blue or black ink. Answers in PENCIL are not marked as it is regarded as rough work.

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QUESTION 1

1.1 1.1.1

$$\ln y = \lim_{x\to 0} \frac{1}{\frac{3\cos 3x}{\sin 3x}}$$

$$\ln y = \lim_{x\to 0} \frac{1}{3}\tan 3x$$

$$\ln y = 0$$

(3)

1.1.2 $y = 1$ (1)

1.2

$$\lim_{x\to\infty} \frac{x^2}{e^x}$$

$$= \lim_{x\to\infty} \frac{2x}{e^x}$$

$$= \lim_{x\to\infty} \frac{2}{e^x} \quad \checkmark$$

$$= \frac{2}{\infty}$$

$$= 0 \quad \checkmark$$

(3)
[7]

QUESTION 2

2.1

$$f(x) = \frac{2-x}{3-2x}$$

$$f(x+h) = \frac{2-(x+h)}{3-2(x+h)} = \frac{2-x-h}{3-2x-2h}$$

$$f'(x) = \lim_{h\to 0}[\frac{2-x-h}{3-2x-2h} - \frac{2-x}{3-2x}].\frac{1}{h}$$

$$f'(x) = \lim_{h\to o} \frac{(2-x-h)(3-2x)-(3-2x-2h)(2-x)}{(3-2x-2h)(3-2x)}.\frac{1}{h}$$

$$f'(x) = \lim_{h\to 0} \frac{h}{(3-2x-2h)(3-2x)}.\frac{1}{h} \quad \checkmark$$

$$f'(x) = \frac{1}{(3-2x)^2} \quad \checkmark$$

(4)

2.2 2.2.1 $y = \dfrac{1-x}{\sqrt{\tan x}}$

$$\frac{dy}{dx} = \frac{-1(\sqrt{\tan x}) - (1-x)\frac{1}{2}(\tan x)^{\frac{-1}{2}}.\sec^2 x}{(\sqrt{\tan x})^2}$$

(4)

2.2.2 $y = [\ln(\cos ec 2x)]^5$

$$\frac{dy}{dx} = 5[\ln(\cos ec 2x)]^4.\frac{-2\cos ec 2x.\cot 2x}{\cos ec 2x}$$

(3)

2.2.3 $y = 2e^{\arcsin 5x}.\sqrt[3]{x}$

✓ ✓

$$y = 2e^{\arcsin 5x}.\frac{5}{\sqrt{1-(5x)^2}}.\sqrt[3]{x} + 2e^{\arcsin 5x}.\frac{1}{3}x^{\frac{-2}{3}}$$

(2)

2.3 $y = (9-x^3)^{\ln x}$

$$\ln y = (\ln x)\ln(9-x^3)$$
$$\frac{1}{y}\frac{dy}{dx} = \frac{1}{x}.\ln(9-x^3) + \ln x.\frac{-3x^2}{(9-x^3)}$$
$$\frac{dy}{dx} = y[\frac{\ln(9-x^3)}{x} + \ln x.\frac{-3x^2}{(9-x^3)}$$ ✓

(4)

2.4 2.4.1 $y = \arcsin\ x$

$$\sin y = x$$
$$\cos y\frac{dy}{dx} = 1$$
$$\frac{dy}{dx} = \frac{1}{\cos y}$$
$$\frac{dy}{dx} = \frac{1}{\sqrt{1-\sin^2 y}}$$ ✓
$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}$$ ✓

(2)

2.4.2 $y = \arcsin(5.10^x)$

$$\frac{dy}{dx} = \frac{5.10^x.\ln 10}{\sqrt{1-(5.10^x)^2}}$$ ✓ ✓

(2)

2.5 $x^2 + 2 = y^3.\cot 3x$

$$2x = 3y^2\frac{dy}{dx}\cot 3x + y^3.-3\cos ec^2 3x$$

$$2x + 3y^3\cos ec^2 3x = 3y^2\cot 3x\frac{dy}{dx}$$

$$\frac{2x + 3y^3\cos ec^2 3x}{3y^2\cot 3x} = \frac{dy}{dx}$$

(4)
[25]

QUESTION 3

3.1 3.1.1

$$f(x) = x(x^2 - 3x - 9)$$
$$f(x) = x^3 - 3x^2 - 9x$$
$$f'(x) = 3x^2 - 6x - 9$$
$$f''(x) = 6x - 6$$
$$6x - 6 = 0$$
$$x = 1$$
$$y = -11$$

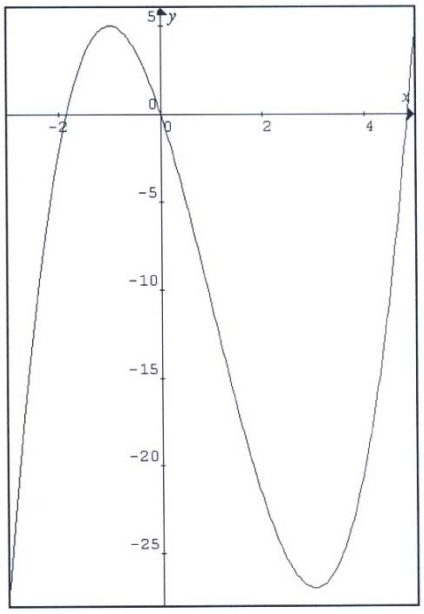
(2)

3.1.2

X	-1	0	1	2	3
Y	5	0	-11	-22	-27

✓✓ (Any 2 correct=1 mark) (2)

3.1.3



✓ – y-intercept
✓ – shape

(2)

3.1.4

$$e = -\frac{f(-0{,}1)}{f'(-0{,}1)} = -\frac{0{,}869}{-8{,}37} = 0{,}104$$ ✓✓
$$a = -0{,}1 + 0{,}104 = 0{,}004$$

(3)

3.2 3.2.1 All these units indicate rates of change ✓ of the volume, radius and height respectively. (2)

3.2.2

$$\frac{dV}{dt} = 2\pi r \frac{dr}{dt}.h + \pi r^2 \frac{dh}{dt}$$ ✓
$$\frac{dV}{dt} = 2\pi(11)(0{,}5)(30) + \pi(11)^2(-1)$$ ✓
$$\frac{dV}{dt} = 656{,}593$$ ✓

(5)

3.3 $D+h=6$

$D=6-h$

$r=\frac{6-h}{2}$

$V=\pi r^2 h$

$V=\pi(3-\frac{h}{2})^2.h$

$V=9\pi h-3\pi h^2+\frac{\pi h^3}{4}$

$V'=9\pi-6\pi h+\frac{3\pi h^2}{4}$

$0=\pi(36-24h+3h^2)$

$0=(h-6)(h-2)$

$h=6;h=2$

$r=0;r=2$

Radius = 2; height = 2

(5)
[21]

QUESTION 4

4.1 4.1.1 $\int\frac{4x}{100-x^2}=\frac{A}{(10-x)}+\frac{B}{(10+x)}dx$

$4x=A(10+x)+B(10-x)$

$x=-10:-2=B$

$x=10:A=2$

(2)

4.1.2 $\int_1^3\frac{4x}{100-x^2}dx$

$\int_1^3\frac{2}{(10-x)}+\frac{-2}{(10+x)}dx$

$=[-2\ln(10-x)-2\ln(10+x)]_1^3$

$=(-9{,}022)-(-9{,}19)$

$=0{,}168$

(3)

4.2 4.2.1 $y = x^{-2}\sin\frac{1}{x}$

$u = \frac{1}{x} \therefore dx = -x^2\ du$

$\int -x^{-2}\ \sin u.x^2\ du$

$= -\int \sin u du$

$= \cos u + c$

$= \cos\frac{1}{x} + c$

4.2.2 $y = \cos^2 5x$

$\int \frac{1}{2} + \frac{1}{2}\cos 10x dx$

$= \frac{1}{2}x + \frac{\sin 10x}{20} + c$

4.2.3 $y = \cos 12x.\sin 3x$

$\int \frac{1}{2}(\sin 15x) - \sin 9x) dx$

$= \frac{1}{2}[\frac{-\cos 15x}{15} + \frac{\cos 9x}{9}] + c$ ✓ ✓

4.2.4 $y = \frac{dx}{\sqrt{25 - 25x^2}}$

$\int \frac{1}{5}\frac{dx}{\sqrt{1 - x^2}}$ ✓✓

$= \frac{1}{5}\sin^{-1} x + c$ ✓

(4 × 3) (12)

4.3 $\int x^{13} \ln 2x \ \ dx$

$f(x) = \ln 2x \qquad g'(x) = x^{13}$

$f'(x) = \frac{1}{x} \qquad g(x) = \frac{x^{14}}{14}$

$\ln 2x.\frac{x^{14}}{14} - \int \frac{1}{x}.\frac{x^{14}}{14} dx$

$= \ln 2x.\frac{x^{14}}{14} - \int \frac{x^{13}}{14} dx$

$= \ln 2x.\frac{x^{14}}{14} - \frac{x^{14}}{196} + c$ ✓✓

(4)
[21]

QUESTION 5

5.1 5.1.1

$\sqrt[3]{x} = \frac{x}{2}$

$x^3 - 8x = 0$

$x = 0; x^2 = 8$

$y = 0; x = \pm 2{,}828$

$y = \pm 1{,}414$

one mark per *x*-value

(3)

5.1.2

$\int_0^{2{,}828} x^{\frac{1}{3}} - \frac{x}{2} dx$

$= [\frac{3}{4} x^{\frac{4}{3}} - \frac{x^2}{4}]_0^{2{,}828}$

$= (1) - (0)$

$= 1$

(4)

5.1.3

$\pi \int_0^{2{,}828} x^{\frac{2}{3}} - \frac{x^2}{4} dx$

$= \pi[\frac{3}{5} x^{\frac{5}{3}} - \frac{x^3}{12}]_0^{2{,}828}$ ✓

$= \pi[(1{,}508) - (0)]$ ✓

$= 4{,}738$

(4)

5.2 $\int_1^2 \frac{2x^2}{5x^3}dx$

$\int_1^2 \frac{2}{5x}dx$

$= [\frac{2}{5}\ln x]_1^2$

$= 0{,}277$ (3)

5.3 $V = \pi\int_0^h \frac{-r}{h}x + r dx$

$= \pi[\frac{-r}{h}.\frac{x^2}{2} + rx]_0^h$

$= \pi[\frac{-r}{h}.\frac{h^2}{2} + rh]$

$= \pi[\frac{-rh}{2} + rh]$

$= \frac{\pi rh}{2}$ (4)

[18]

QUESTION 6

6.1 $e^x dx - y^3 dy = e^{-x}dx$

$dx(e^x - e^{-x}) = y^3 dy$ ✓

$e^x + e^{-x} + c = \frac{y^4}{4}$ ✓ (4)

6.2 $(1+x)\sin y.dy = \cos y.dx$

$\tan y dy = \frac{1}{1+x}dx$

$\ln\sec y = \ln(1+x) + c$ (4)

[8]

TOTAL: 100

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